Solving Percent Problems with Proportions

You can also write percent problems as proportion problems. One ratio compares the percent to 100. The other ratio compares the part to the whole. Use a variable to represent the missing element. Then solve the proportion.

Example 1

Hide

Nick got an 84% on his last test. If there were 50 questions on the test, how many questions did he answer correctly?

Step 1

Set up a proportion.

$$\frac{\text{percent}}{100} = \frac{\text{part}}{\text{whole}} \qquad \frac{84}{100} = \frac{p}{50} \rightarrow (50)\left(\frac{84}{100}\right) = \left(\frac{p}{50}\right)(50) \qquad 42 = p$$

Step 2

Solve the proportion.

$100 \times p = 84 \times 50$

$100 \times p = 4{,}200$

$p = 4{,}200 \div 100$

$p = \mathbf{42}$

Nick correctly answered **42** out of 50 questions.

Example 2

Hide

In a recent election, the mayor received 3,600 votes, or 45% of the vote. How many people voted in the election?

Step 1

Set up a proportion.

$$\frac{\text{percent}}{100} = \frac{\text{part}}{\text{whole}} \qquad \frac{45}{100} = \frac{3{,}600}{w} \rightarrow (w)\left(\frac{45}{100}\right) = \left(\frac{3600}{w}\right)(w)$$

$$\frac{45w}{100} = 3600 \qquad \frac{(0.45)w}{0.45} = \frac{3600}{0.45} \qquad w = 8000$$

Step 2

Solve the proportion.

$45 \times w = 100 \times 3{,}600$

$45 \times w = 360{,}000$

$w = 360{,}000 \div 45$

$w = \mathbf{8{,}000}$

8,000 people voted in the election.

Example 3

Hide

If 7 out of 20 students bring their lunch to school, what percent of students bring their lunch?

Step 1

Set up a proportion.

$$\frac{\text{percent}}{100} = \frac{\text{part}}{\text{whole}} \qquad \frac{n}{100} = \frac{7}{20} \rightarrow (100)\left(\frac{n}{100}\right) = \left(\frac{7}{20}\right)(100)$$

$$n = \frac{700}{20} \qquad n = 35$$

Step 2

Solve the proportion.

$20 \times n = 7 \times 100$

$20 \times n = 700$

$n = 700 \div 20$

$n = \mathbf{35}$

35% of students bring their lunch to school.

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Lesson 7: Percentages

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